Crypto Investigator

Overview and Quick Start — v0.7.0, 2026-07-10 17:50 UTC

What it does

A cryptocurrency tracing and warrant-preparation tool for law enforcement. Starting from a victim's wallet, it follows stolen or scammed funds across the public blockchain until they reach a business - usually a crypto exchange - that can be served with a subpoena or warrant for the account holder's records. It then packages everything needed to draft that legal process: the deposit addresses, funding transactions, timestamps, amounts (with approximate USD values at the transaction dates), and draft records-request language.

It does not identify people. It identifies where legal process can be served; the human behind an account comes from the exchange's KYC records via that process. It never guesses: everything that is an inference rather than an on-chain fact is labelled with a confidence level and its basis, and dead ends (mixers, privacy coins, smart contracts) are reported honestly.

How it works, at 30,000 feet

1. The investigator creates a case and enters the victim's wallet address - and, if known, the transaction hash of the specific payment to the scammer, so only that payment is followed.

2. The tool pulls public blockchain records and follows the money hop by hop (breadth-first, up to a chosen depth), recording every address and movement.

3. Each address it reaches is checked against the official OFAC sanctions list and a list of known exchange wallets, and classified: pass-through, exchange (an exit point), sanctioned, mixer, smart contract, or busy service.

4. Results appear as a left-to-right money-flow map (victim on the left, exits on the right) with plain-language explanations on every element, plus ranked exit-point cards.

5. One click produces the court-ready PDF report, the chain-of-custody CSV, the raw JSON, and an FBI IC3 complaint worksheet prefilled from the trace.

Every single network request the tool makes is logged with its exact URL, UTC time and a SHA-256 hash of the response, so an independent analyst can re-run and verify any piece of the work. A separate Technical Validation Document describes the methodology in full for expert review.

Running the program

Requirements: Windows/macOS/Linux with Python 3.10 or newer and a modern browser. Internet access for live tracing (previously fetched records are served from the local evidence cache).

Install (once): from the program folder, run `python -m pip install -r requirements.txt`

Start: run `python run.py` and open `http://127.0.0.1:8321` in a browser. The app runs entirely on the local machine; nothing is exposed to the network and case data never leaves the computer.

First-time setup (Settings button): paste a free Etherscan API key (needed for Ethereum tracing only; Bitcoin needs no key) and click 'Download / update OFAC sanctions list' once, then periodically.

Workflow: Step 1 create/pick a case → Step 2 paste the victim's wallet address → Step 3 (optional) paste the transaction hash of the payment to investigate → Step 4 choose how many hops to follow → press 'Follow the money' → review the exit points and map → download the report, custody log and IC3 worksheet.

What comes out

- Court-ready fund-tracing report (PDF) with methodology, limitations and chain-of-custody appendices.
- Chain-of-custody log (CSV) - every data pull, re-runnable.
- Raw trace data (JSON) for archival or other tools.
- FBI IC3 complaint worksheet (PDF) mirroring the official form at complaint.ic3.gov, prefilled with the traced payments.

Where to look next

- **Technical Validation Document** (docs/validation/) - full methodology, integrity controls, and step-by-step verification procedures for outside experts.
- **CHANGELOG.md** - what each version added and its known limitations. **docs/listings/** - the complete source code of every version in a single hashed file.